Unit 2 - Computer Systems
Assignment 8 - Managing Upgrades (Part 2)

**Introduction**

**Scenario/Vocational Context**: *You are a Junior Technician in Jersey Bank and have been asked to review the reasons for upgrades within the Bank. You have been asked to prepare a document.*

**Task 1**

**(U29:M3) - Design and implement a procedure to preserve data integrity during an upgrade**

**Task description:** *The Finance Manager is concerned about the integrity of their data before, during and after the upgrade process.*
*You are required to create a set of procedures that are designed to preserve the data integrity during an upgrade from Windows 8.1 to Windows 10*

*These procedures will need to be explained and must cover at least the following areas: copy of registry data, copy of user data and prior images of the system.*

*The procedure will need to be demonstrated and you will complete a work record, an observation record will be completed by your tutor*

Description of procedures: (Resources required, problems associated with the procedure).

| **2.8 Task 1 Planning Grid** | **Description of what is involved with the task.** | **What resources would be required to carry out the task** | **What could happen if these tasks were not carried out at the start of the installation/upgrade** |
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| Copy of Registry data | When creating a copy of registry data, the user can't just backup and restore the registry like any other file.<br><br>Much of it can't be modified by hand, and the files themselves can't be replaces or copied over, at least not | The resources that is needed to create a copy of the registry data would be an external hard drive, USB flash drive and IT staff.<br><br>For the external hard drive, this will be used to store the entirety of the registry data, which will | If these tasks were not carried out at the start of the upgrade, there will be a risk that the upgrade would make the registry files corrupted which can be due to the OS change from Windows 8.1 to Windows 10, making them unable to be accessed. |

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| | while Windows is running. The user also can't restore a full backup of the registry from an export file either.<br><br>There are a number of things the the user can do to backup sections of the registry, and the user can also use the System Restore to actually restore the registry back to a previous state, but that counts for the whole system, instead of the registry data itself<br><br>When copying and restoring sections of the registry, the user will have to find the files within the registry editor itself, and export the selected file in any preferred location in the pc, such as the desktop.<br><br>When Backing up the whole registry to a .reg file, the user can choose the "Computer" tab and export the entire registry onto a preferred location on the pc, which is exactly like the exportation of sections. | be the main source of backup if there was a problem with the upgrade or other technical difficulties which would cause the upgrade to not go to plan.<br><br>For the USB flash drive, this would be used as a secondary storage of the backup of the registry data. This is just incase the external hard drive has corrupted files, or any other forms of malfunctions.<br><br>For the IT staff, they will need to make sure to make this backup and the upgrade in a time where the staff or other users aren't using the computers, just incase something goes wrong with the backup or upgrade process.<br><br>For example, during the weekend, holidays or the hours where the staff on the computer systems stop working and leave the workplace.<br><br>The reason for this is that it would interrupt working hours, which could cause the business to lose time, making them lose on profits. | For instance, if the computer system is upgraded to Windows 10, and the upgrade failed, causing the Registry values to no longer be available for use, it could be very problematic.<br><br>This problem would mean that additional work would need to be done to recreate the system configuration, causing time and money to be wasted within the business, due to the extensive amount of time needed. |
| Copy of user Data | When creating a copy of user data, the user would need to backup all the personal files, | Just like the previous method, the resources that is needed to create a copy of the user data | If these tasks were not carried out at the start of the upgrade, there will be a risk that the |

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| | folders and data in a way that can be easily read and used in Windows 10, from Windows 8.1.<br><br>To do this, the user will have to calculate how much space is required, locate the important data files and finally copy the data to an external drive.<br><br>When calculating how much space is required, the user would need to check the size of the internal hard drive, external hard drive and the USB flash drive.<br><br>When locating important data files, the user will need to find important data files located outside of the user folder, such as Outlook data files, WIndows Mail emails and data, MYOB company files and backups and Quicken company files.<br><br>When copying the data to an external drive, the user will need to open and send the necessary files on the external hard drive and USB drive, checking that they were transferred. | would be an external hard drive, USB flash drive and IT staff.<br><br>For the external hard drive, this will be used to store the entirety of the user data, which will be the main source of backup if there was a problem with the upgrade or other technical difficulties which would cause the upgrade to not go to plan.<br><br>For the USB flash drive, this would be used as a secondary storage of the backup of the user data. This is just incase the external hard drive has corrupted files, or any other forms of malfunctions.<br><br>For the IT staff, they will need to make sure to make this backup and the upgrade in a time where the staff or other users aren't using the computers, just incase something goes wrong with the backup or upgrade process.<br><br>For example, during the weekend, holidays or the hours where the staff on the computer systems stop working and leave the workplace.<br><br>The reason for this is that it would interrupt working hours, which could cause | upgrade would make some files corrupted due to the OS change, making them unable to be used.<br><br>For instance, if the computer system is upgraded to Windows 10, and the upgrade failed, making the files corrupted or even deleting them, it could cause a wide amount of problems.<br><br>One of the problems is that that additional work would need to be done to recreate the system configuration, causing time and money to be wasted within the business, due to the extensive amount of time needed.<br><br>Another problem is that the files that were corrupted or deleted may have contained valuable information for the company, meaning that all the data would either have to be recreated or, in the worst case scenario, never be obtained again. Causing the company to be in a bigger disadvantage. |

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| | | the business to lose time, making them lose on profits. | |
| Prior image of system | When creating a prior image of the system, the user would have to do a simple step by step process.<br>However, it differs for each Operating System, so this will be an example of Windows 8.1 since the upgrade is from Windows 8.1 to Windows 10.<br><br>In this process the user would have to open the control panel to navigate to the System and security tab.<br><br>Which the user would then open the file history, which would link the user to the System Image Backup at the bottom-left corner of the window.<br><br>The user can then restore these backup images using the Advanced Startup Options on Windows 8.1, which can be accessed from a windows installation disc or a recovery drive. | Exactly like the previous methods, the resources that is needed to create an image of the system would be an external hard drive, USB flash drive and IT staff.<br><br>For the external hard drive, this will be used to store the system image, which will be the main source of backup if there was a problem with the upgrade or other technical difficulties which would cause the upgrade to not go to plan.<br><br>For the USB flash drive, this would be used as a secondary storage of the backup of the system image. This is just incase the external hard drive has corrupted files, or any other forms of malfunctions.<br><br>For the IT staff, they will need to make sure to make this backup and the upgrade in a time where the staff or other users aren't using the computers, just incase something goes wrong with the backup or upgrade process.<br><br>For example, during the weekend, holidays or the hours where the staff on | Exactly like the copy of user data, if these tasks were not carried out at the start of the upgrade, there will be a risk that the upgrade would make some files corrupted due to the OS change, making them unable to be used.<br><br>For instance, if the computer system is upgraded to Windows 10, and the upgrade failed, making the files corrupted or even deleting them, it could cause a wide amount of problems.<br><br>One of the problems is that that additional work would need to be done to recreate the system configuration, causing time and money to be wasted within the business, due to the extensive amount of time needed.<br><br>Another problem is that the files that were corrupted or deleted may have contained valuable information for the company, meaning that all the data would either have to be recreated or, in the worst case scenario, never be obtained again. Causing |

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| | | the computer systems stop working and leave the workplace.<br><br>The reason for this is that it would interrupt working hours, which could cause the business to lose time, making them lose on profits. | the company to be in a bigger disadvantage. |

Summary of how these tasks would be added to a plan for an upgrade

In conclusion, the whole point of having data integrity is to save the user from the hassle of a computer system or a company's system being corrupted during an upgrade. Giving the user the ability to revert back to the previous version of the system before it was upgraded, saving the user time and money, which may be limited.

Basically, these tasks would be good to have as a plan for an upgrade and should be added. To add these tasks to a plan for an upgrade, the user would have to find the window of time where the staff and other users aren't using the computer system for working. Meaning that the upgrade wouldn't be able to interrupt the use of the computer system, which also means that time and money wouldn't be wasted.

Overall, I believe that having these tasks added to a plan for an upgrade would be very useful due to the minimisation of risks during the upgrade, which can prove to be very problematic. To add these plans, the manager will basically need to find a time where the staff and other people aren't working on the computer system and would have enough time to backup the computer systems without it interfering with working time.

Also, the reason that the resources needed are the same is due to the fact that this is all done within the same computer system, with the same OS and the same business. So no other resources are needed in order to make a recovery point for the computer systems within the business.

<u>Demonstration of a procedure to preserve data integrity</u>

**Company Name:** Jersey Bank PLC
**Contact:** **IT Manager**
**Company Address: 9 Hill Street, St Helier**
**Company Phone:** **00 44 1534 678456**

**Work Record**

Summary of task: Implement a procedure to preserve data integrity during an upgrade (User Data)

Assignment Number: 2.8 Technician: Leandro Baroca

Task Number: 1 Criteria Number: 29 M3

Environment: (Software): Windows 10

User Platform: (Hardware): x64

Detail of work carried out:

1. Firstly, before starting the installation I need to decide when the best time for it to take place in order to keep the business open as usual or minimise downtime. I did this by checking the times that the staff are working, and checking if the computer systems aren't being used by any member of staff for business use.

2. Secondly, I would have to plug in my USB or external hard drive. Then I will have to open my user folder: Computer > System (C:) > Users > [username], which I then should now see all the user folders (Contacts, Desktop, Downloads, etc). Starting at the top of the list, right-click on the first folder, then I mouse-over the Send to: menu item.

3. Thirdly, I clicked on the USB or external hard drive from the list, it will usually be named the same as it's brand, for example: WD_Elements (E:) or Sandisk (F:). By then, Windows automatically copied the selected folder and it's contents to the backup drive. I then repeated this process for each folder but I had to be aware, since it can take some time (hours) to transfer large amounts of data (ie. My Documents or My Music) especially to USB flash drives or external hard drives.

4. Finally, when I finished transferring all the user folders, copy over any extra data files to your backup drive to (Outlook, Quicken etc). Then all the backup of all the user data was complete, and I was able to carry out the upgrade to Windows 10 with the backup ready just incase there is a malfunction.

**Task 2**

**(U29:M4) - Design a procedure to back out of software upgrades**

**Task description:** *Your manager wants to be reassured in the unlikely that the whole or parts of the upgrade is not successful.*

*You need to add to create procedures that will allow you to "back out" of software upgrades. These procedures will need to be explained and: must cover at least the following areas: changing the operating system back to Windows 8.1, system recovery points and rolling back device drivers.*

Description of procedures: (Resources required, problems associated with the procedure)

| **2.8 Task 2 Planning Grid** | **Description of what is involved with the task.** | **What resources would be required to carry out the task** | **What would be involved to use the facility to "back-out" of the upgrade.** | **If this utility did not work what would be needed to be done to restore the system** |
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| System recovery points | When creating a system recovery point, the user would have to do a simple step by step process.<br><br>However, it differs for each Operating System, so this will be an example of Windows 8.1 since the upgrade is from Windows 8.1 to Windows 10.<br><br>The user would have to navigate the control panel into system and security, where the user would go into System Protection.<br><br>Within the System Properties dialog box, the user would click the System Protection tab and | The resources that is needed to create a system recovery point would be an internal hard drive and IT staff.<br><br>For the internal hard drive, this will be used to store the system recovery point, which will be the main source of backup.<br><br>If there was a problem with the upgrade or other technical difficulties which would cause the upgrade to not go to plan.<br><br>For the IT staff, they will need to make sure to make this recovery point and the upgrade in a time where the staff or other users aren't using the computers, | How would you use a system recovery point to restore a system?<br><br>To use the system recovery point, the use would first need to log onto the computer as an administrator.<br><br>The user would then need to navigate through the control panel, onto System and Security and onto System protection.<br><br>Then the user will click System Restore, with the recommended restore being the default choice. This restore is the most recent restore point and the best place to start. | If this utility failed to work, the user would still have different choices to restore the system.<br><br>Choices such as restore backups on external hard drives, cloud backups or in the worst case scenario, installing a new system.<br><br>For the backups on external hard drives, the user can use the external hard drive to transfer the copy of the previous system onto the computer, meaning that no previous data is lost.<br><br>For backups on the cloud, it's somewhat the same process as the external hard drive, but it is |

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| | then click the Create button.<br><br>Which the user would then make a name for the restoration point and creates it, windows would then give a message saying that it was created successfully.<br><br>Meaning that it can be used for its intended purpose. | just incase something goes wrong with the upgrade process.<br><br>For example, during the weekend, holidays or the hours where the staff on the computer systems stop working and leave the workplace.<br><br>The reason for this is that it would interrupt working hours, which could cause the business to lose time, making them lose on profits. | However, the user can choose a different restore point, by clicking “Choose a different restore point”, click Next and then click a restore point.<br><br>When the user clicked Next, Finish to confirm it and then clicked Yes, the system restore would be complete and computer will restart. | downloaded instead of connected with wiring. Which means that this form of backup also means that no previous data is lost.<br><br>With reinstalling a new system, this should be done if the previous methods did not work. This method is when the user reinstalls a new system into the computer, which does mean the previous files are gone and need to be remade.<br><br>However, it is still better than a computer without a system. |
| Rolling back device drivers | When rolling back a device driver, this feature is used to uninstall the current driver for a hardware device and then automatically install the previously installed driver.<br><br>Basically, rolling back a driver as a quick and easy way to uninstall the latest driver, and then reinstall the previous one, all in one simple step. | Like the previous method, the resources that is needed to rollback the device drivers would be an internal hard drive and IT staff.<br><br>For the internal hard drive, this will be used to store the system recovery point, which will be the main source of backup.<br><br>If there was a problem with the upgrade or other technical difficulties | To use the rollback of the device driver, the user will have to open the Device Manager. Doing so via Control Panel would be the easiest method.<br><br>In the Device Manager, the user will locate the device that the user would want to roll back the driver for.<br><br>After finding the hardware the user is rolling back the | If this utility failed to work, the user would still have different choices to restore the system.<br><br>Choices such as restore backups on external hard drives, cloud backups or in the worst case scenario, installing a new system.<br><br>For the backups on external hard drives, the user can use the external hard drive to transfer the copy |

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| | This process is the same no matter what driver the user needs to roll back, whether it be an NVIDIA video card driver, advanced mouse/keyboard driver, etc.<br><br>Rolling back a driver in Windows usually takes as long as 10 minutes or more depending on the driver and what hardware it's for. | which would cause the upgrade to not go to plan.<br><br>For the IT staff, they will need to make sure to make this rollback and the upgrade in a time where the staff or other users aren't using the computers, just incase something goes wrong with the upgrade process.<br><br>For example, during the weekend, holidays or the hours where the staff on the computer systems stop working and leave the workplace.<br><br>The reason for this is that it would interrupt working hours, which could cause the business to lose time, making them lose on profits. | driver for, the user will right click on the device's name and choose Properties.<br><br>In the Properties window for the device, the user will click on the Driver tab.<br><br>From the Driver tab, the user will click the Roll Back Driver button.<br><br>Then the user would have to do some conformations by clicking Yes, and then the computer would ask for a restart.<br><br>After the user click yes, the computer will automatically restart, and it will load with the previous device driver. | of the previous system onto the computer, meaning that no previous data is lost.<br><br>For backups on the cloud, it's somewhat the same process as the external hard drive, but it is downloaded instead of connected with wiring. Which means that this form of backup also means that no previous data is lost.<br><br>With reinstalling a new system, this should be done if the previous methods did not work. This method is when the user reinstalls a new system into the computer, which does mean the previous files are gone and need to be remade.<br><br>However, it is still better than a computer without a system. |
| Summary of how these tasks would be added to a plan for an upgrade<br><br>In conclusion, like the backup of data, the whole point of having a recovery point is to save the user from the hassle of a computer system or a company's computer system being corrupted or failing during an upgrade. Giving the user the ability to revert back to the previous version of the system before it was upgraded.<br><br>Basically, adding these methods to a plan for an upgrade, the user would have to carry this out if the | | | | |

upgrade of a computer system doesn't go to plan.

So that it would require the user to cancel the upgrade, and without a point of return for the computer system, the data the computer system last had would all be lost, causing the company to lose valuable information and time.

Overall, I believe that having these tasks added to a plan for an upgrade would be very useful due to the minimisation of risks of the company losing valuable data and time, which can prove to be very problematic.

To add these plans, the manager will need to have a backup plan if the upgrade during the time where the staff isn't working on the computer, so that nothing will be lost of all goes wrong. If the upgrade went wrong without the backup plan, the computer systems would all be unusable, wasting data, time and profit for the business.

Also, the reason that the resources needed are the same is due to the fact that this is all done within the same computer system, with the same OS and the same business. So no other resources are needed in order to make a recovery point for the computer systems within the business.